UCF Center for Research in Computer Vision

CAP 6412 Advanced Computer Vision

Term Project Status Presentation

Group Sensor-VLM

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Presenter: Rafael Fernandes da Silva

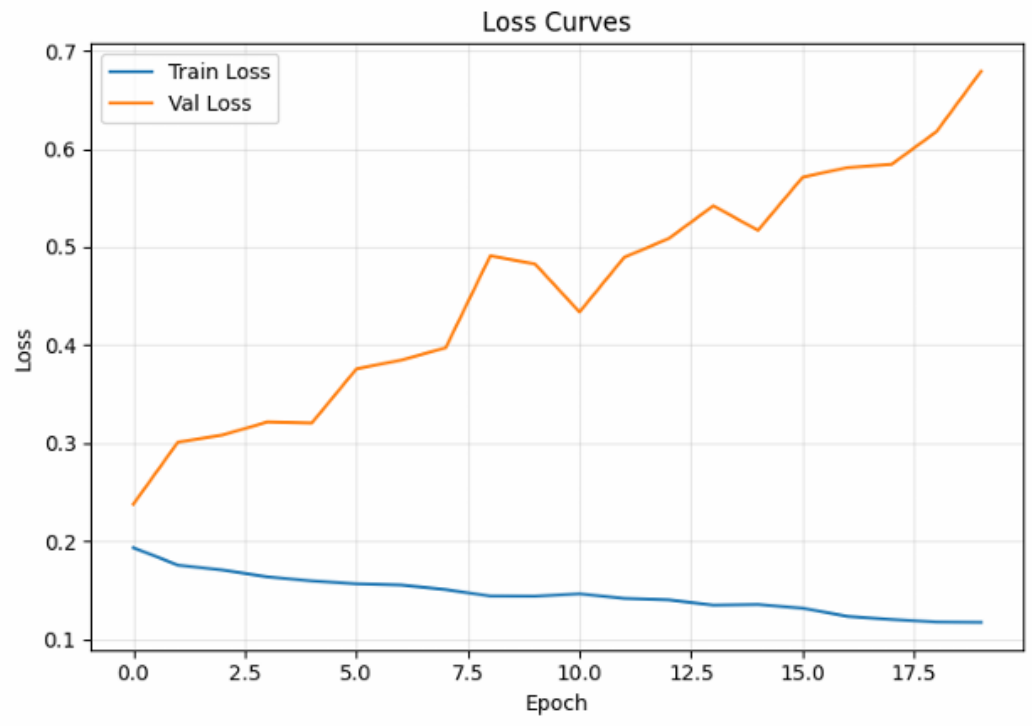
Recap

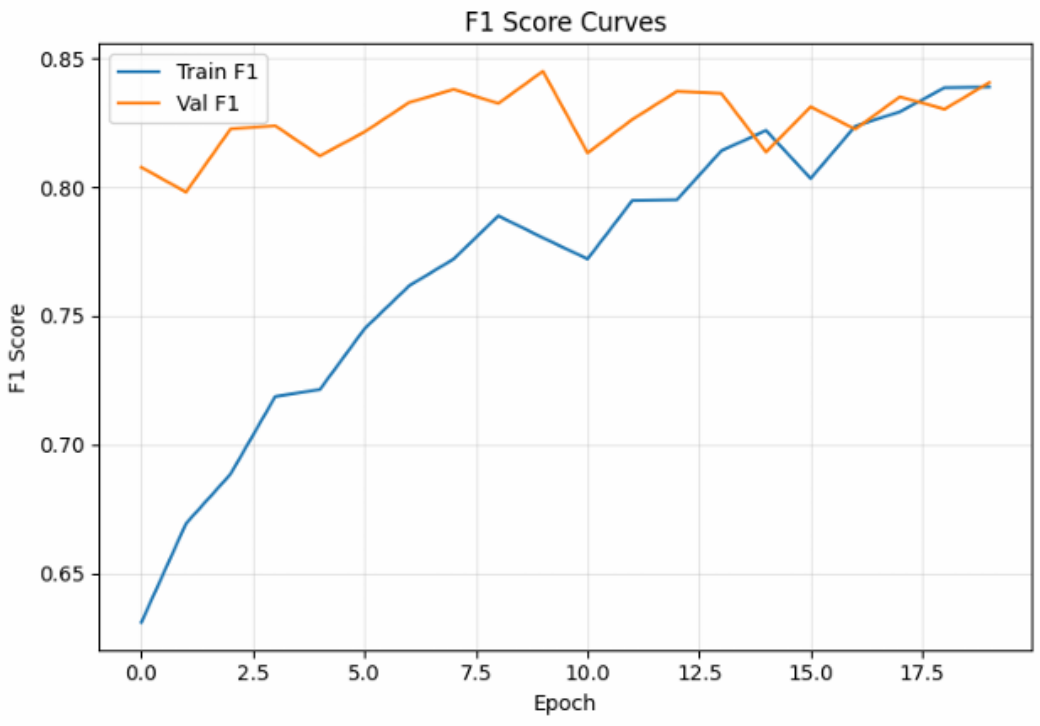
- Identifying Ambiguity in Vision Language Instructions
- BLIP-2 Pipeline
- MLP Classifier
  - Binary
  - Types of Ambiguity
- Clarification Question:
    - If ambiguity detected:
    - Template-based using scene

MLP Training Results

- Accuracy: 0.7881
- F1 Score: 0.8778
- Precision: 0.8645
- Recall: 0.8916
- Loss: 0:3300
- 4574 unique instructions from DialFRED
  - 86/14
  - 384, ~130k params

Training Results Visualized





BLIP-2 Results

- Backbone ready
  - Producing multimodal embeddings
  - To be connected to the MLP
  - Multi-caption variance
  - VQA probing

Issues

- Dataset imbalance
  - 86% ambiguous
  - Fine tuning
  - TEACh, ALFRED
- Text-only embedding
  - BLIP-2 Embeddings to MLP

Plan to Complete

- Embeddings from BLIP-2 to MLP
- Clarification question system
- Classify the type of Ambiguity

Contributions

- MLP Training – Rafael, Shoshanah, Obee
- Research & Presentation – Sky
- BLIP-2 Pipeline - Erick